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ABSTRACT:

This project presents a GUI-based Smart Calculator developed using Python and Tkinter. The calculator performs basic arithmetic operations along with additional functions like square and square root. It includes keyboard input support and a history feature to store previous calculations. The project focuses on enhancing user interaction through a modern graphical interface.

INTRODUCTION:

A calculator is an essential tool used for performing mathematical operations quickly and accurately. Traditional command-line programs are less interactive, so this project focuses on creating a Graphical User Interface (GUI) based calculator. The aim is to provide a user-friendly experience with improved design and functionality.

OBJECTIVES:

- **To design a calculator using Python**
- **To implement GUI using Tkinter**
- **To perform arithmetic operations**
- **To add square and square root functions**
- **To implement history using file handling**
- **To improve user interface with modern design**

TOOLS & TECHNOLOGIES:

- **Python**
- **Tkinter (GUI library)**
- **VS Code / IDLE**

SYSTEM WORKING:

The calculator takes input through buttons or keyboard. The entered expression is evaluated using Python functions, and the result is displayed on the screen. Each calculation is stored in a file named "history.txt", which can be accessed using the History button.

CODE SCREENSHOTS:

ADVANTAGES:

- **Easy to use**
- **Fast calculations**
- **User-friendly interface**
- **Stores previous results**
- **Supports keyboard input**

LIMITATIONS:

- **Limited to basic operations**
- **No advanced scientific functions**
- **UI limited by Tkinter**

FUTURE SCOPE:

- **Add scientific functions (sin, cos, log)**
- **Improve UI using advanced libraries**
- **Convert into mobile/web application**

CONCLUSION:

The Smart Calculator project successfully demonstrates the implementation of a GUI-based application using Python. It performs mathematical operations efficiently while providing additional features like history and keyboard support. This project helped in understanding GUI design and file handling concepts effectively.